

AI Critique

I used Claude Sonnet 5 to generate the initial JMeter plans, analyse the Load, Stress and Spike JTL files, and create a reusable performance-testing skill. The first output was structurally convincing but incomplete. It encoded the wrong login failure action, allowed product creation after a failed catalog read, used a Transaction Controller that added synthetic samples, parsed the growing product response too heavily, and accepted an empty product array. These mistakes could have changed the measured throughput or allowed an invalid business journey to appear successful. In Task 2, the numerical analysis was much stronger, but the recommendation to use 70 users as normal concurrency was unsupported because only 20 users were sustained in the formal Load and Endurance runs. The Stress test passed through approximately 70 users only briefly during its ramp.

The AI missed these issues because it produced plausible JMeter structures and generalized from common performance-testing heuristics without fully validating JMeter's serialized action values, controller sample behavior, data growth, or the difference between a short ramp interval and a stable operating point. I verified and corrected the work by reading the API and backend source, comparing the original and corrected JMX files, checking every JTL label and response code, recalculating endpoint percentiles and 30-second buckets, and comparing them with each dashboard's `statistics.json`. I also preserved the Stress aggregate discrepancy instead of choosing one value silently.

The final skill was useful because it converts these corrections into repeatable safeguards. My main lesson is that AI should generate reviewable hypotheses and drafts, while human judgment must protect measurement validity. A performance conclusion is acceptable only when its workflow, calculation, execution evidence and limitations can all be traced to real artifacts.